

Critical Thinking

CLAIMS & ARGUMENTS I



THE PLAN!

- ❑ What is Critical Thinking?
- ❑ Beliefs and Claims, Fact and Opinion
 - ❑ In-Class Exercises
- ❑ What is an Argument?
 - ❑ In-Class Exercises

WHAT IS CRITICAL THINKING?



REFLECT & DISCUSS

Q1: What is critical thinking?

- Explain

WHAT IS CRITICAL THINKING?

Critical Thinking:

- The systematic evaluation or formulation of beliefs or statements by rational standards.
- This is about the *quality* of your beliefs.

WHAT IS CRITICAL THINKING?

Critical Thinking:

- is *systematic*: it involves distinct procedures and methods.
- entails *evaluation* (used to assess existing beliefs) and *formulation* (used to devise new beliefs).
- operates according to *rational standards*: beliefs are judged by how well they are supported by *reasons*.

WHAT IS CRITICAL THINKING?

Critical Thinking

- We all want to hold *well-supported* beliefs and avoid *unsupported (or false)* ones.
 - Practical reason: beliefs inform our actions.
 - A well-supported belief has *good* reasons supporting it.

Can you name a source of *good*
reasons for belief?

Direct Observation



Reliable Sources



Good Arguments

- 1. If it's raining outside, then it's cloudy. (True)
- 2. It's not cloudy. (True)
- Conclusion: Therefore, it's not raining. (True)

WHAT IS CRITICAL THINKING?

- Critical thinking is *not directly* about what you think, but about *how* you think.
 - Critical thinking is not directly concerned with investigating what caused you to have the beliefs you do.
 - For example, you may hold certain moral views because they align with the society in which you live...
 - Or, you may hold some beliefs because you are emotionally attached to them...

W

- Critical thinking: how you think
- Critical thinking: how you think
- For example, how you think about society
- Or, you can think about the world

Darius, a king of ancient Persia, was intrigued by the variety of cultures he encountered in his travels. He had found, for example, that the Callatians (a tribe of Indians) customarily ate the bodies of their dead fathers. The Greeks, of course, did not do that—the Greeks practiced cremation and regarded the funeral pyre as the natural and fitting way to dispose of the dead. Darius thought that a sophisticated understanding of the world must include an appreciation of such differences between cultures. One day, to teach this lesson, he summoned some Greeks who happened to be present at his court and asked them what they would take to eat the bodies of their dead fathers. They were shocked, as Darius knew they would be, and replied that no amount of money could persuade them to do such a thing. Then Darius called in some Callatians, and while the Greeks listened asked them what they would take to burn their dead fathers' bodies. The Callatians were horrified and told Darius not even to mention such a dreadful thing.

KING?

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Herodotus
484 BC – 425 BC

WHAT IS CRITICAL THINKING?

- Critical thinking is not about *what* you do in fact believe, but about what you *should* or *ought* to believe.

WHAT IS CRITICAL THINKING?

- A central (obvious?) tenet:
 - (Tenet) A belief is worth believing if we have *good reasons* to accept it. The better the reasons, the more likely the belief is true, and the more it is worth believing.
 - Another way is to say that *the purpose of thinking critically is to come to correct conclusion.*

WHAT IS CRITICAL THINKING?

- Sounds pretty simple, right?
- The idea: to provide a set of skills and tools to help you think critically
 - Translation: to help you evaluate your own and others' beliefs according to rational standards to determine whether those beliefs are worth holding onto.
 - Or as our text puts it, critical thinking is thinking that critiques. It helps you critique *reasoning*, broadly construed – the thinking used in arriving at decisions, developing plans, coming to conclusions, offering hypotheses, and so forth.

WHAT IS CRITICAL THINKING?

- One thing we will focus on are common errors of reasoning.
- Psychologists call *very common* (or *nearly universal*) errors *cognitive biases*.

WHAT IS CRITICAL THINKING?

- One thing we will focus on are common errors of reasoning.
- Psychologists call *very common* (or *nearly universal*) errors *cognitive biases*.
- These can lead to false conclusions and bad decisions.
 - For example, we tend to judge facts differently depending on how they are stated or framed...

WHAT IS CRITICAL THINKING?

- Two ways of describing the same state of affairs, framed differently:
 - (1) A company is *saving* 300 jobs (out of 1000)
 - (2) A company is *laying off* 700 people (out of 1000)



WHAT IS CRITICAL THINKING?

- Two ways of describing the same state of affairs, framed differently:
 - (1) A company is *saving* 300 jobs (out of 1000)
 - (2) A company is *laying off* 700 people (out of 1000)
- People often think of (1) positively and (2) negatively, despite the result being the same. The implication: people are more sympathetic to (1).
- This is a nearly universal *cognitive bias*.

WHAT IS CRITICAL THINKING?

- Critical thinking also involves *logic*.

WHAT IS CRITICAL THINKING?

- Critical thinking also involves *logic*.
 - The study of good reasoning, or inference, and the rules that govern it.
 - From the Greek *logos*, meaning “word” or “reason” or “logic.”

WHAT IS CRITICAL THINKING?

- For example:
 - An invalid ("bad") argument:
 - 1. If it's raining outside, then it's cloudy.
 - 2. It's not raining outside.
 - 3. Therefore, it's not cloudy.
 - A valid ("good," if sound) argument:
 - 1. If it's raining outside, then it's cloudy.
 - 2. It's not cloudy.
 - 3. Therefore, it's not raining.

WHAT IS CRITICAL THINKING?

- We will focus on the structure of arguments this and next class.

WHY DOES IT MATTER?

- Our lives are defined by our actions and choices, and our actions and choices are defined by our thoughts and beliefs.

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- Our lives are defined by our actions and choices, and our actions and choices are defined by our thoughts and beliefs.
 - Why do you choose to eat certain foods?
 - Why do you turn left when you leave the front door?
 - Why do you come to class on time? (At all?)

WHY DOES IT MATTER?

- Our lives are defined by our actions and our thoughts



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WHY DOES IT MATTER?

- Our lives are defined by our actions and choices, and our actions and choices are defined by our thoughts and beliefs.
 - Why do you choose to eat certain foods?
 - Why do you turn left when you leave the front door?
 - Why do you come to class on time? (At all?)
- So, being critical about your beliefs can help you understand what actions and choices make sense for you.

Critical Thinking, the Long Version

The Collegiate Learning Assessment (CLA) Project of the Council for Aid to Education has come up with a list of skills that covers almost everything your authors believe is important in critical thinking. If you achieve mastery over all these or even a significant majority of them, you'll be well ahead of most of your peers—and your fellow citizens. In question form, here is what the council came up with:

How well does the student

- determine what information is or is not pertinent;
- distinguish between rational claims and emotional ones;
- separate fact from opinion;
- recognize the ways in which evidence might be limited or compromised;
- spot deception and holes in the arguments of others;
- present his/her own analysis of the data or information;
- recognize logical flaws in arguments;
- draw connections between discrete sources of data and information;
- attend to contradictory, inadequate, or ambiguous information;
- construct cogent arguments rooted in data rather than opinion;
- select the strongest set of supporting data;
- avoid overstated conclusions;
- identify holes in the evidence and suggest additional information to collect;
- recognize that a problem may have no clear answer or single solution;
- propose other options and weigh them in the decision;
- consider all stakeholders or affected parties in suggesting a course of action;
- articulate the argument and the context for that argument;
- correctly and precisely use evidence to defend the argument;
- logically and cohesively organize the argument;
- avoid extraneous elements in an argument's development;
- present evidence in an order that contributes to a persuasive argument?

MATTER?

- The goal is to help you to use critical thinking skills instinctively.

HOW IT WORKS

- We will begin with some of the *basic concepts* of critical thinking and work out from there.

BELIEFS AND CLAIMS, FACT AND OPINION



REFLECT & DISCUSS

*Q2: What are the differences between the following terms:
belief, judgment, opinion, statement, claim, assertion*

BELIEFS AND CLAIMS, FACT AND OPINION

Belief (or Judgment or Opinion):

- When we come to a conclusion, we have a belief (or a judgment or an opinion).
 - *When a belief (judgment, opinion) is asserted in a declarative sentence, the result is a claim or statement.*

BELIEFS AND CLAIMS, FACT AND OPINION

Note that the text uses beliefs, judgments, and opinions interchangeably

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BELIEFS AND CLAIMS, FACT AND OPINION

Belief (or Judgment or Opinion):

- When we come to a conclusion, we have a belief (or a judgment or an opinion).
 - *When a belief (judgment, opinion) is asserted in a declarative sentence, the result is a claim or statement.*
- **Example:** If you conclude that the battery of your car is dead, you *believe* that the car battery is dead.
 - "The car battery *is* dead."
 - In other words, you make a claim (statement or assertion) that something *is* or *is not* the case.

BELIEFS AND CLAIMS, FACT AND OPINION

Statement (or Claim)

- An assertion that something is or is not the case.
- Examples:
 - "Taxes are too high."
 - "The earth is not flat."
 - "Neither of us wants ice cream."
 - "A triangle has three sides."
 - "I am cold."
 - "The capital of Pakistan is Karachi."

BELIEFS AND CLAIMS, FACT AND OPINION

Statement (or Claim)

- Statements can be *true* or *false* (sometimes hard to tell which).
 - Some of the list were false.
- They can be *positive* (asserting a claim: "You are a liar.") or *negative* (denying a claim: "You are not a liar.").

BELIEFS AND CLAIMS, FACT AND OPINION

- Beliefs and claims are *propositional*: they can be expressed in true-or-false declarative sentences.

BELIEFS AND CLAIMS, FACT AND OPINION



REFLECT & DISCUSS

Q3: Are all opinions subjective? Are all factual claims true?

- *What is an objective claim?*
- *What is an subjective claim?*

BELIEFS AND CLAIMS, FACT AND OPINION

- An *objective* claim has this characteristic: Whether it is true or false is independent of whether you or anyone else thinks it is true or false.
 - “There is life on Jupiter” is an objective claim.
 - *Why?*
 - Because whether or not life exists there doesn’t depend on whether you (or anyone else) thinks it does.

BELIEFS AND CLAIMS, FACT AND OPINION

- An *objective* claim has this characteristic: Whether it is true or false is independent of whether you or anyone else thinks it is true or false.
 - Put differently: An *objective* statement is a statement that is not made true or false by the speaker or writer's thinking it is true or false.

BELIEFS AND CLAIMS, FACT AND OPINION

- An *objective* claim has this characteristic: Whether it is true or false is independent of whether you or anyone else thinks it is true or false.
 - Although *objective* claims are either true or false, we may not know which a given claim is.

BELIEFS AND CLAIMS, FACT AND OPINION

- An *objective* claim has this characteristic: Whether it is true or false is independent of whether you or anyone else thinks it is true or false.
 - Although *objective* claims are either true or false, we may not know which a given claim is.
 - *Example:*
 - “Portland, Oregon, is closer to the North Pole than the equator” is true.
 - “Portland, Oregon, is closer to the equator than to the North Pole” is false.
 - “More stamp collectors live in Portland, Oregon, than in Portland Maine” is an objective claim whose truth or falsity is not known, at least not by us.

BELIEFS AND CLAIMS, FACT AND OPINION

- Not every declarative sentence expresses an **objective** claim:
 - Judgments, such as “Bruno Mars has swag,” are **subjective**.
 - *Why?*
 - Because whether a **subjective** statement is true or false is *not independent* of whether you think it is true or false.
 - Or put it differently, a **subjective** statement is a statement that is made true or false by the speaker or writer’s thinking it is true or false.
 - Another central tenet of critical thinking: your believing something does not make it true.

BELIEFS AND CLAIMS, FACT AND OPINION

- Not every declarative sentence expresses an **objective** claim:
 - Judgments, such as “Bruno Mars has swag,” are **subjective**.
 - *Why?*
 - Because whether a **subjective** statement is true or false is *not independent* of whether you think it is true or false.
 - On the contrary, *a **subjective** judgment about something is true if you think it is true (you can't be mistaken).*
 - Because a **subjective** judgment cannot be viewed as probably true or as probably false, it isn't the sort of thing that can be thought of as supportable by evidence.
 - “The strawberry tastes great!”

BELIEFS AND CLAIMS, FACT AND OPINION

- In short, **subjective** judgments:
 - cannot be mistaken;
 - are not subject to probability quantifications;
 - are not the sort of thing for which evidence could be given,
 - should not be invoked to dismiss any particular statement as unworthy of discussion.

BELIEFS AND CLAIMS, FACT AND OPINION

- Sometimes people talk about the difference between *fact* and *opinion*, having in mind that all *opinions* are *subjective*.

BELIEFS AND CLAIMS, FACT AND OPINION

- Sometimes people talk about the difference between *fact* and *opinion*, having in mind that all *opinions* are *subjective*.
- But some opinions are not *subjective* judgments, because their truth or falsity is independent of what people think.
- Example:
 - If you believe that Portland, Oregon, is closer to the North Pole than to the equator, that opinion happens to be true, and would continue to be true even if you change your mind.

BELIEFS AND CLAIMS, FACT AND OPINION

- Sometimes people talk about the difference between *fact* and *opinion*, having in mind that all *opinions* are *subjective*.
- But some opinions are not *subjective* judgments, because their truth or falsity is independent of what people think.
- We can refer to objective opinions as *factual* opinions or beliefs, but that *does not mean that factual opinions are all true*.
 - Example:
 - "Portland, Oregon, is closer to the equator than to the North Pole" is a factual opinion that is false.

BELIEFS AND CLAIMS, FACT AND OPINION

- Sometimes people talk about the difference between *fact* and *opinion*, having in mind that all *opinions* are *subjective*.
- But some opinions are not *subjective* judgments, because their truth or falsity is independent of what people think.
- We can refer to objective opinions as *factual* opinions or beliefs, but that *does not mean that factual opinions are all true*.
 - A *factual* opinion/belief/claim = an *objective* opinion/belief/claim = an opinion/belief/claim whose truth is independent of whether anyone thinks it is true.

IN-CLASS EXERCISES

Q4: 4 Would you classify the following claims as objective or subjective?

- a) On a baseball field, the centre of the pitcher's mound is 59 feet from home plate.
- b) Green is the most pleasant colour to look at.
- c) Driving while drowsy is dangerous.
- d) Pit vipers can strike a warm-blooded animal even when it is pitch dark.
- e) Donald Trump is overweight
- f) This lemon tastes great because it isn't bitter.
- g) There is life on Mars.
- h) Your teacher will complain if you text in class.
- i) Your teacher would be crazy not to complain if you text in class.
- j) God exists.
- k) Donald Trump is fat.

Q4: 4 Would you classify the following claims as objective or subjective?

- a) On a baseball field, the centre of the pitcher's mound is 59 feet from home plate. (Obj.)
- b) Green is the most pleasant colour to look at. (Subj.)
- c) Driving while drowsy is dangerous. (Obj.)
- d) Pit vipers can strike a warm-blooded animal even when it is pitch dark. (Obj.)
- e) Donald Trump is overweight (Obj.)
- f) This lemon tastes great because it isn't bitter. (Subj.)
- g) There is life on Mars. (Obj.)
- h) Your teacher will complain if you text in class. (Obj.)
- i) Your teacher would be crazy not to complain if you text in class. (Subj.)
- j) God exists (Obj.)
- k) Donald Trump is fat (Subj.)

- Read the passage “Determining the Kind of Claim You Should Make” (worksheet)
- **Q5:** *How might you justify the conceptual claims mentioned in the passage?*

- Read the passage “Determining the Kind of Claim You Should Make” (worksheet)
- *Q5: How might you justify the conceptual claims mentioned in the passage?*
 - 1. For **claims of fact or existence**, you must provide evidence that a situation is, in fact, as you characterize it.
 - 2. **Claims of definition or classification** depend on reasoning about similarities or differences that assigns an entity to some broader class or distinguishes it from other entities.
 - 3. Finally, **claims of cause** or **consequence** connect sets of facts to show that some situation does (or doesn't) follow from or lead to another.
 - 4. **Effective claims of evaluation** or **appraisal** depend on criteria of judgment to justify why something is good or bad (or better or worse than something else)